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Mood states recognition based on Mandarin speech and deep learning in patients with bipolar disorder

Jinhao Li^{1,2†}, Qiuping Yan^{1,3†}, Minghui Li^{1†}, Xiaofei Hou¹, Chao Li⁴, Ziping Zhao⁴, Huifang Yin^{1*} and Guangming Xu^{1*}

Abstract

Background Current assessments of bipolar disorder mainly rely on clinician's experience and patients' self-reporting, which has the risk of observer bias and subjective cognitive bias. There is an urgent need for more objective indicators to improve the accuracy of recognizing and monitoring mood states in patients with bipolar disorder. Speech might be a promising marker for recognizing different mood states of bipolar disorder, because of its accessible, objective, and non-invasive features. This study aims to construct a deep learning model based on Mandarin speech to recognize depressive episode, (hypo)manic episode, and euthymic state in bipolar disorder.

Methods Fifty-three patients with bipolar disorder were recruited from the outpatient clinic in Tianjin Anding Hospital from March 2023 to April 2024. Speech recordings were collected in a controlled acoustic environment and were pre-processed. Three speech features including the Mel spectrogram, HuBERT and WavLM self-supervised feature were extracted. The speech feature with the highest recognition rate was selected as the basic speech feature for the mood states recognition. Three mood states recognition models were constructed based on deep learning algorithms: (1) the LSTM model, (2) the ECAPA-TDNN model, and (3) our proposed model (Ours). The Ours model enhances the ECAPA-TDNN architecture by integrating an LSTM layer to capture contextual information, followed by processing through a TFA mechanism. The recognition performance of the constructed models is evaluated by WA, UA, and Macro_F1 value.

Results 19 patients with bipolar disorder were in a (hypo)manic episode, 15 in a depressive episode, and 19 in a euthymic state. The WavLM self-supervised feature had the highest speech recognition rate (90.13%), followed by the HuBERT self-supervised feature (67.56%) and the Mel Spectrum feature (58.09%). The Ours model achieved the best overall performance, with UA, WA, and Macro_F1 of 85.94%, 85.79%, and 85.78%, respectively, which is 6–8% higher than those of LSTM and ECAPA-TDNN.

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Conclusions This study validates the effectiveness of the fusion model of self-supervised features and deep learning in Mandarin speech recognition of bipolar disorder patients' mood states. The findings provide evidence for the use of speech features as biomarkers to accurately distinguish mood states in bipolar disorder.

Clinical trial number Not applicable.

Keywords Bipolar disorder, Mood states, Mandarin speech, Self-supervised features, Deep learning

Introduction

Bipolar disorder is a severe mental illness characterized by recurrent episodes of abnormal mood states, affecting more than 1% of the global population [1]. As one of the most disabling mental disorders [2, 3], bipolar disorder is associated with an increased risk of suicide and substantially impaired social functioning [4–6]. In clinical practice, pharmacological interventions constitute the primary therapeutic approach for bipolar disorder [7]. Treatment regimens must be implemented promptly and individualized according to the each patient's specific clinical presentation and treatment efficacy depends on the accurate identification and management of pathological affective states. However, early and accurate recognition of affective states in bipolar disorder remains a major challenge in clinical settings [8]. Psychiatrists typically use either standardized diagnostic tools or psychopathology rating scales, complemented by clinical judgment or patient self-report. These traditional assessment methods are vulnerable to observer bias and subjective cognitive bias [9, 10], particularly during (hypo)manic episode when patients' insight of the illness may be diminished [11]. Consequently, there is an urgent need for more objective indicators to improve the accuracy of recognizing and monitoring mood states in patients with bipolar disorder.

Clinical observations suggest that patients with bipolar disorder in a (hypo)manic state exhibit a higher pitch and increased speech rate, while those in a depressive state tend to show the opposite pattern [12, 13]. Such mood-related variations in speech indicate that speech features may be crucial for assessing symptom severity in bipolar disorder. Recent studies show that voice data represent a promising direction for the objective measurement of emotional states [14, 15]. For example, analyses of speech recorded during text reading can automatically predict depressive states [16]. Another study using a smartphone application to extract voice features from six patients with bipolar I disorder, demonstrated that changes in voice data could detect depressive and hypomanic symptoms [17]. Therefore, speech features are promising for differentiating mood states in bipolar disorder because they are accessible, objective, and non-invasive.

However, research on speech features in psychiatry has predominantly focuses on major depressive disorder or schizophrenia [18, 19], with relatively fewer studies in

bipolar disorder. Moreover, these studies primarily utilize speech parameters to distinguish bipolar disorder from other populations [20–22], rather than to differentiate between distinct mood states. Additionally, many prior studies were conducted in English-speaking contexts [12, 17], limiting the direct transferability of findings to other languages. In particular, English is an intonation language, whereas Chinese is a tonal language. They differ in phonological systems and acoustic features [23, 24]. For example, mandarin Chinese distinguishes semantic meanings through tonal variation, so pitch convey both semantic and emotional information [25], increasing the complexity of emotion recognition in Chinese. Studies have also found that key acoustic features, such as fundamental frequency (F0) and tone stress value (dB), differ between Mandarin and English [24]. These acoustic features function are important sources of information for speech emotion recognition [26]. Due to these differences in acoustic features, both feature selection strategies and weight allocation need to be optimized for Chinese when constructing emotion recognition models. Conducting research on mood state recognition in bipolar disorder based on Mandarin speech is therefore of substantial significance for advancing in this field.

Deep learning offers a powerful paradigm in speech emotion recognition. Compared with traditional machine learning algorithms, deep learning can capture complex structures and features without the need for manual extraction and adjustment. It can also extract low-level features from raw data and process unlabeled data [27]. Multiple neural architectures have shown strong performance; for example, recurrent architectures such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) perform excellently in natural language processing and speech classification [28]. Previous studies have reported high success in audio recognition tasks [29], and have significantly improved performance in depression detection using deep learning [30]. Building on this progress, the present study explores deep learning-based speech analysis to enhance mood state recognition in bipolar disorder.

Given the strong performance of self-supervised learning in speech signal processing [31, 32], this study employs self-supervised feature extraction to characterize mood states in bipolar disorder within the Mandarin language context. Through comparisons with traditional

spectral features, such as Mel spectrograms, features with optimal recognition efficiency are identified and adopted as core inputs for speech mood state recognition. Furthermore, cutting-edge deep learning algorithms are adopted to construct a recognition model for the three mood states ((hypo)manic episode, depressive episode, and euthymic state) of bipolar disorder based on Mandarin speech. The findings provide a new research paradigm for speech recognition research of bipolar disorder in the Mandarin language context, and offer a methodological foundation and empirical evidence to advance precise, multilingual speech-based assessment of psychiatric disorders.

Methods

Participants

This study used a cross-sectional design and the research process is summarized in Fig. 1. Participants were recruited from the psychiatric outpatient clinic at Tianjin Anding Hospital between March 2023 and April 2024. Inclusion criteria comprised: (1) aged between 18 and 65 years old; (2) diagnosed with bipolar disorder (type I or II) according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) criteria [33]; (3). attained primary school education or higher and capable of completing the study; (4) provided informed consent and agreed to participate in the study. Exclusion criteria included (1) comorbid organic brain diseases, severe physical illnesses, or intellectual developmental disorders; (2) pregnancy; (3) and current DSM-5 diagnosis of another mental disorder.

Clinical assessments

Participants were interviewed by trained psychiatrists using the Structured Clinical Interview for DSM-5 Disorders, Research Version (SCID-5-RV) [34]. Information on demographic and clinical characteristic was also collected.

The severity of symptoms and functional impairment was assessed using the 17-item Hamilton Depression Rating Scale (HAMD-17) [35], the Bech-Rafaelsen Mania Scale (BRMS) [36] and the Functioning Assessment Short Test Scale (FAST) [37], respectively. After the assessments, participants were divided into three groups based on HAMD-17 and BRMS scores: (1) depressive episode group (HAMD-17 score ≥ 17 and BRMS score ≤ 5); (2) (hypo)manic episode group (BRMS score > 5 and HAMD-17 score < 17); and (3) euthymic state group (HAMD-17 score < 17 and BRMS score ≤ 5).

Speech sample collection

Speech recordings were collected in a controlled acoustic environment: a soundproofed room with electromagnetic shielding to eliminate interference. Audio data were

acquired using an iFLYTEK Smart Recorder (SR502) configured for lossless WAV format. Participants maintained a fixed 20 cm distance from the microphone to ensure optimal signal. The recordings consisted of structured clinical interviews between trained clinicians and participants, incorporating standardized psychiatric assessments (SCID-5-RV, HAMD-17, BRMS, and FAST) [34–37]. Throughout recording, storage, and transmission, data protection procedures were strictly followed to safeguard participant privacy.

Speech signal preprocessing

The collected patient speech samples were divided into 3-second segments. The final dataset comprised 2990 audio samples, including 1041 for (hypo)manic episodes, 1028 for depressive episodes, and 921 for speech during euthymic states. Analyses were conducted in Python using Visual Studio Code (version 1.78.0). To ensure uniform data distribution, the dataset was shuffled 500 times. Subsequently, 80% of the samples were allocated to the training set and 20% to the test set via randomized stratification. Reported results represent the average across five independent resampling iterations to enhance reliability.

Speech feature extraction

Both Mel-spectrogram features [38] and self-supervised speech representations (HuBERT, WavLM) [31, 38] were utilized and extracted using Python. While Mel-spectrograms are well-established features for speech recognition [39], self-supervised approaches have demonstrated superior performance across multiple speech processing tasks [32].

Constructing speech recognition model of mood states in bipolar disorder

Three comparative models were implemented: (1) The Long Short-Term Memory Network (LSTM) [40], a class of recurrent architectures that address short-term memory limitations in standard RNNs and are widely used in speech recognition [41, 42]. (2) The Emphasised Channel Attention, Propagation, and Aggregation in Time-Delay Neural Networks (ECAPA-TDNN) [43], which have shown strong performance in speech emotion recognition [44]. (3) The Ours model (Fig. 2): Adding LSTM to the ECAPA-TDNN model before SE_Res2Block to capture contextual information, then pass in SE_Res2Block to enrich the mood state representation, and finally pass in Time-Frequency Attention (TFA) [45] to obtain the timestamps associated with mood state detection and the frequency bands.

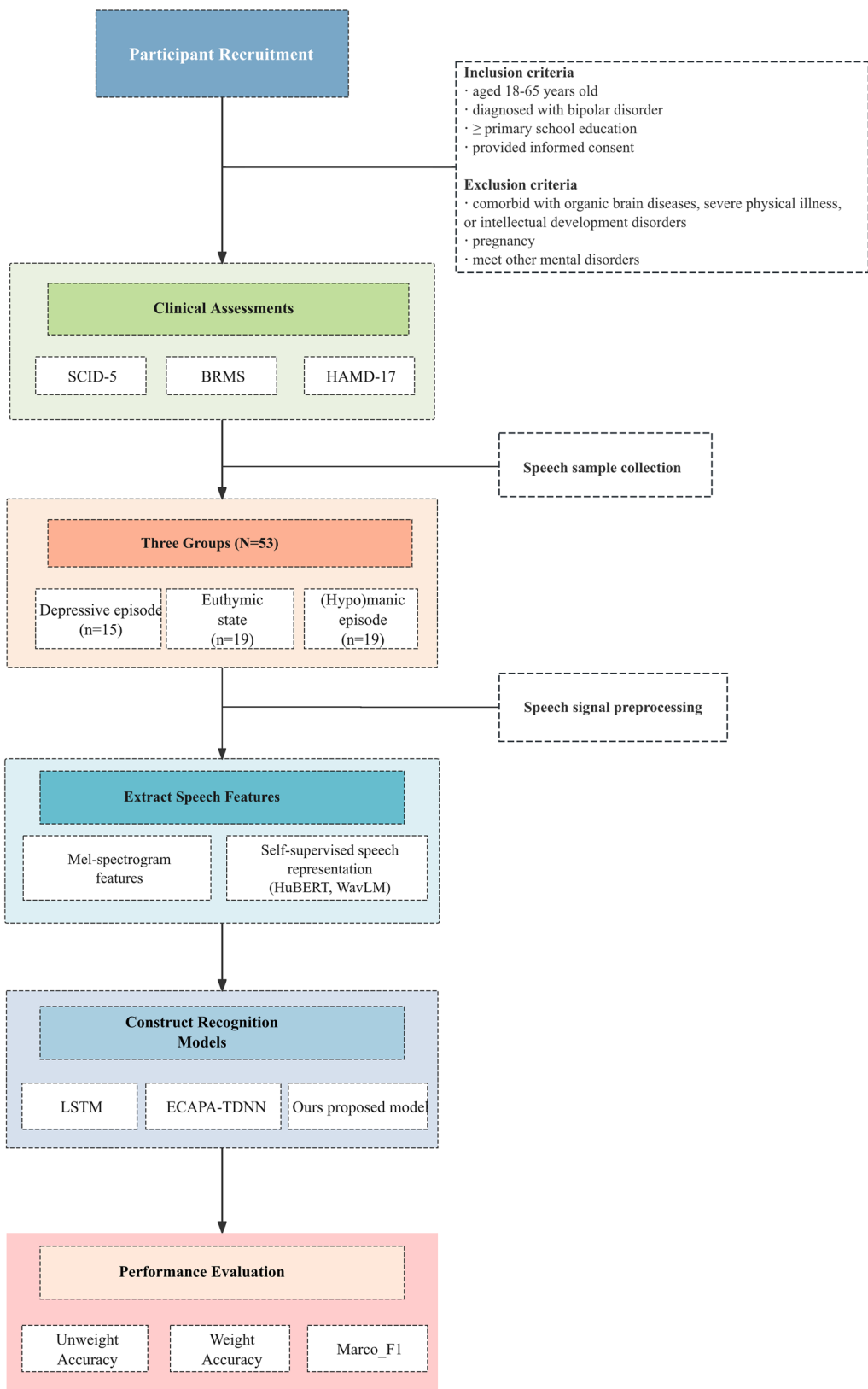


Fig. 1 The overall workflow of the study. Note: SCID-5, Structured Clinical Interview for Diagnostic and Statistical Manual of Mental Disorder, Fifth Edition; BRMS, Bech-Rafaelsen Mania Rating Scale; HAMD-17, 17-item Hamilton Depression Rating Scale; LSTM, Long Short-Term Memory; ECAPA-TNDD, Emphasized Channel Attention, Propagation and Aggregation in TDNN

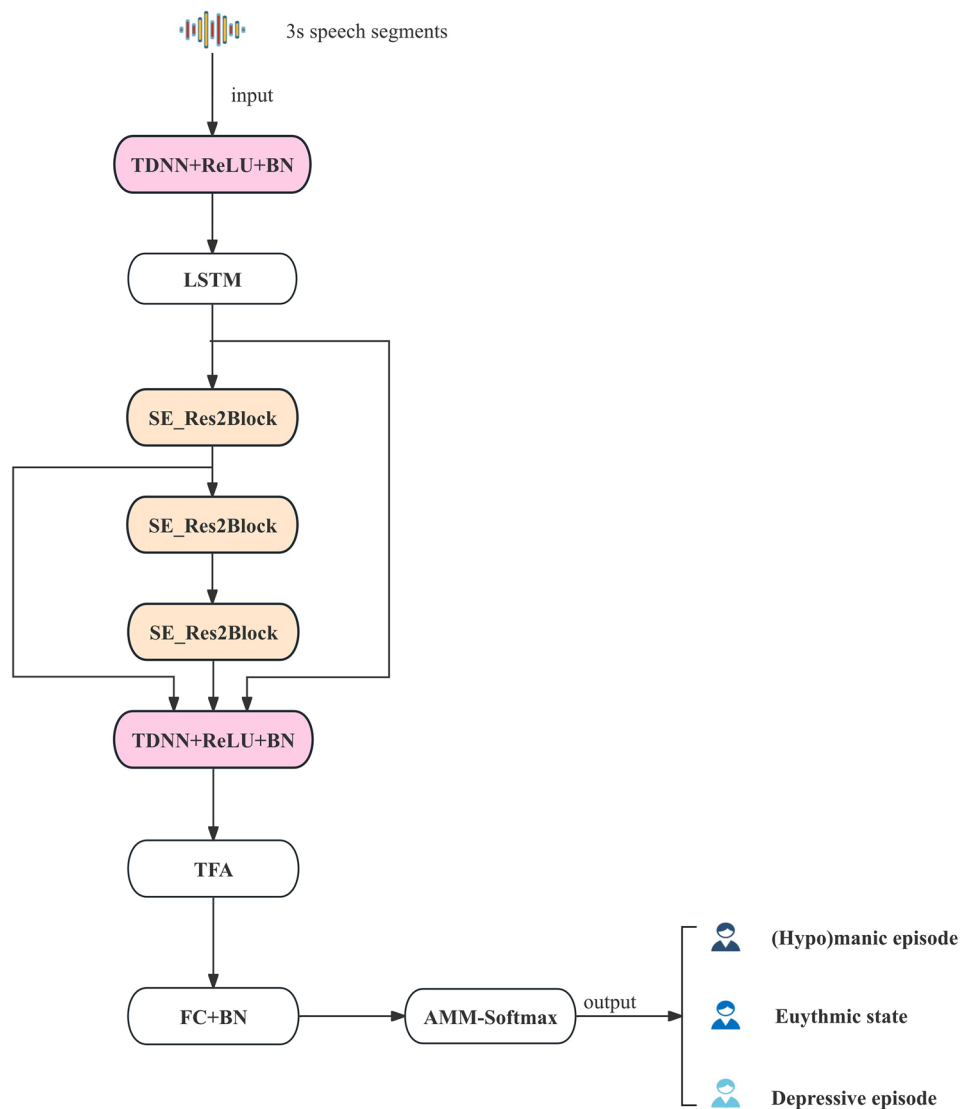


Fig. 2 Ours model schematic. Note: TDNN, Temporal Delay Neural Network; ReLU: Rectified Linear Unit; Batch Normalization; SE-Res2Block, Squeeze-and-Excitation Res2 Block; TFA, Time-Frequency Attention; FU, Fully Connected; AMM-softmax: Angular Margin Metric Softmax

Statistical analysis

Demographic data and clinical characteristics of the three groups were analysed using SPSS 24.0. Continuous variables were tested for normal distribution of data using the K-S test. And non-normally distributed continuous variables were tested using the Mann-Whitney U test (non-parametric) and reported as Median (M) and Inter quartile range (IQR). For categorical variables, the chi-square test was used and results were expressed as a percentage and frequency [% (n)].

This study implemented stratified random sampling to split the dataset into training (80%) and test (20%) sets, ensuring that the proportion of each mood state was preserved across subsets. Three metrics were selected to evaluate model performance comprehensively: (1) Weighted Accuracy (WA), which accounts for class

distribution and reflects overall performance in setting where mood state prevalence may vary. (2) Unweighted Accuracy (UA), which treats all classes equally and helps avoid bias toward majority classes (critical for the smaller depressive episode group). (3) And Macro_F1, which balances precision and recall across all classes and provides a robust measure for the performance in multi-class classification (three mood states) while mitigating the impact of class imbalance.

Results

Comparison of socio-demographic and clinical characteristics of patients with bipolar disorder

The study included 53 patients with bipolar disorder, who were categorized into three groups based on their current mood states: (hypo)manic episode ($n=19$), depressive

Table 1 Socio-demographic and clinical characteristics of patients with bipolar disorder

Variables	Total (n = 53)	(Hypo)manic episode (n = 19)	Euthymic state (n = 19)	Depressive episode (n = 15)	H/ χ^2	p
Age ^a	35.0 (23.0,49.5)	40.0 (33.0,51.0)	34.0 (22.0,54.0)	25.0 (19.0,38.0)	8.238	0.016
Female, %(n)	37.7(20)	42.1(8)	31.6(6)	40.0(6)	0.494	0.781
Full-time job, %(n)	45.3(24)	52.6(10)	42.1(8)	40.0(6)	18.711	0.044
Years of education, %(n)					15.275	0.227
≤ 9	18.9(10)	31.6(6)	5.3(1)	20.0(3)		
10–12	18.9(10)	10.5(2)	26.3(5)	20.0(3)		
> 12	62.2(33)	57.9(11)	68.4(13)	60.0(9)		
Marital status, %(n)					9.726	0.137
Married	49.1(26)	68.4(13)	47.4(9)	26.7(4)		
Unmarried	43.4(23)	26.3(5)	47.4(9)	60.0(9)		
Divorced	5.7(3)	0.0(0)	5.3(1)	13.3(2)		
Widowed	1.8(1)	5.3(1)	0.0(0)	0.0(0)		
HAMD ^a	5.0(1.5,20.5)	2.0(0.0,4.0)	4.0(1.0,5.0)	23.0 (22.0,24.0)	33.396	< 0.001
BRMS ^a	3.0(0.5,16.5)	18.0(16.0,20.0)	1.0(0.0,3.0)	1.0(0.0,2.0)	36.789	< 0.001
FAST ^a	13.0 (5.0,21.5)	13.0 (5.0,22.0)	7.0 (1.0,19.0)	14.0 (13.0,31.0)	7.535	0.023
Age of onset (of disease) ^a	24.0 (18.0,33.5)	25.0 (20.0,36.0)	21.0 (18.0,32.0)	21.0 (16.0,29.0)	1.856	0.395
Illness duration (year) ^a	7.0 (2.0,17.5)	10.0 (1.0,21.0)	9.0 (4.0,19.0)	5.0 (1.0,9.0)	3.946	0.139
Number of Depressive episode ,%(n)					14.123	0.028
≤ 4	58.5(31)	52.6(10)	73.7(14)	46.7(7)		
5–10	20.8(11)	36.8(7)	15.8(3)	6.7(1)		
11–20	9.4(5)	10.6(2)	0.0(0)	20.0(3)		
> 20	11.3(6)	0.0(0)	10.5(2)	26.7(4)		
Number of (Hypo)manic episode, %(n)					4.637	0.591
≤ 4	64.2(34)	47.4(9)	68.4(13)	80.0(12)		
5–10	18.9(10)	26.3(5)	15.8(3)	13.3(2)		
11–20	13.2(7)	21.0(4)	10.5(2)	6.7(1)		
> 20	3.7(2)	5.3(1)	5.3(1)	0.0(0)		

Note: ^a represents continuous variables: expressed as median (Inter Quartile Range, IQR) with a statistic of H

episode ($n = 15$), and euthymic state ($n = 19$). The demographic and clinical characteristics of the participants are presented in Table 1 (at the end of the document text file). Among all the participants, the median age was 35.0 years (IQR: 23.0–49.5), the median illness duration was 7.0 years (IQR: 2.0–17.5), and the median age at onset was 24.0 years (IQR: 18.0–33.5). Females ($n = 20$) comprised 37.7% of the sample.

The analysis revealed that the depressive episode group participants had a significantly younger age and higher FAST score than the (hypo)manic episode group and the euthymic state group. The percentage of participants with a full-time job in the depressive episode group was also lower than that of the other two groups. There were no statistically significant differences in gender, years of education, marital status, age of onset, illness duration, or number of (hypo) manic episodes between groups.

Table 2 Comparison of speech features parameters in patients with bipolar disorder(%)

Speech Feature Parameters	Accuracy	Sensitivity	Specificity	F1 score
MEL spectral	58.09	57.19	78.99	56.20
HuBERT	67.56	66.65	83.65	63.67
WavLM	90.13	90.10	95.10	90.02

Comparison of speech features parameters in patients with bipolar disorder

As shown in Table 2, the MEL spectral features achieved an accuracy of 58.09%, sensitivity of 57.19%, specificity of 78.99%, and an F1 score of 56.20%. HuBERT demonstrated an accuracy of 67.56%, sensitivity of 66.65%, specificity of 83.65%, and an F1 score of 63.67%. Notably, WavLM demonstrated the highest performance across all metrics, achieving an accuracy of 90.13%, sensitivity of 90.10%, specificity of 95.10%, and an F1 score of 90.02%.

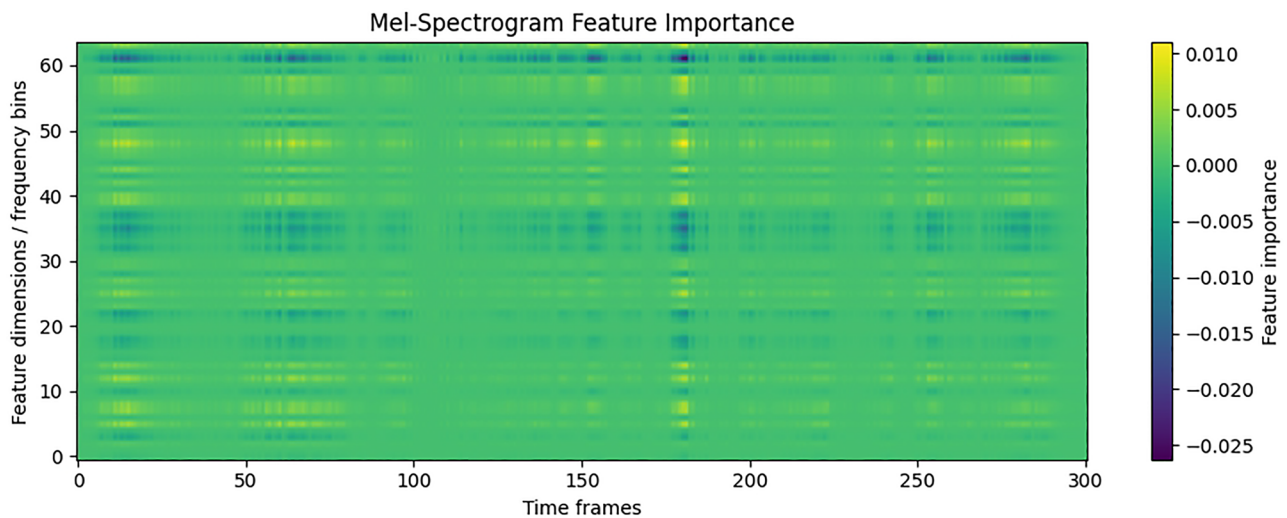


Fig. 3 Feature importance of mel-spectrogram across time frames and frequency bins

Table 3 Comparison of identification results between ours model and baseline model (%)

Models	UA	WA	Macro_F1
LSTM	79.07	79.00	79.08
ECAPA-TDNN	83.39	83.41	83.42
Ours	85.94	85.79	85.78

Figure 3 shows the feature importance of Mel spectrum for model prediction. The colour intensity indicates the magnitude of contribution, with positive values (yellow/green) representing positive contributions and negative values (blue/purple) representing negative contributions. Notably, specific frequency bins (e.g., around 50 and 170–180 time frames) demonstrate relatively high positive importance, indicating these temporal-frequency regions contribute more significantly to the target task. In contrast, the region around 170 time frames and upper frequency bins shows marked negative importance, implying these features may inhibit predictive performance.

Performance evaluation of a speech recognition model for mood states in bipolar disorder

The recognition performance of the LSTM, ECAPA-TDNN, and the Ours models is presented in this section. To further validate the effectiveness of our model, two sets of comparative experiments were conducted. First, the Ours model was compared with the baseline models (LSTM and ECAPA-TDNN). Second, an ablation study [46] was performed by removing the TFA layer from the Ours model (denoted as -TFA+Ours) and comparing its performance with that of the complete model.

To provide a more detailed analysis, the confusion matrices of all models are presented in Figs. 2–5. These matrices supplement the quantitative results in Tables

Table 4 Results of ablation study (%)

Models	UA	WA	Macro_F1
(-)TFA+Ours	83.69	83.48	83.48
Ours	85.94	85.79	85.78

3 and 4 by visualizing the correspondence between predicted and actual values, allowing for a more intuitive evaluation of model performance.

WA, UA, and Macro_F1

This study evaluated the performance of the Ours model proposed in comparison with two baseline models (LSTM and ECAPA-TDNN) for mood state recognition in bipolar disorder, as presented in Table 3. Compared with LSTM (UA 79.07%, WA 79.00%, Macro_F1 79.08%) and ECAPA-TDNN (UA 83.39%, WA 83.41%, Macro_F1 83.42%), the Ours model achieved the highest performance (UA 85.94%, WA 85.79%, Macro_F1 85.78%).

Table 4 presents the ablation study results, where the (-)TFA+Ours model achieves UA, WA, and Macro-F1 scores of 83.69%, 83.48%, and 83.48%, respectively. The Ours model demonstrates significant improvements over this ablated version, with respective increases of 2.25% in UA, 2.31% in WA, and 2.30% in Macro_F1 scores.

Comparison of models confusion matrices

Figure 4 presents the confusion matrix for the LSTM model, revealing distinct recognition patterns across mood states. The model demonstrates stronger performance in identifying euthymic state (84.48%) and depressive episode (82.38%) compared to (hypo) manic episode (72.90%). Notably, the manic episode shows the lowest accuracy, with over 25% of samples being misclassified as either a depressive episode or a euthymic state. As depicted in Fig. 5, the confusion matrix of the

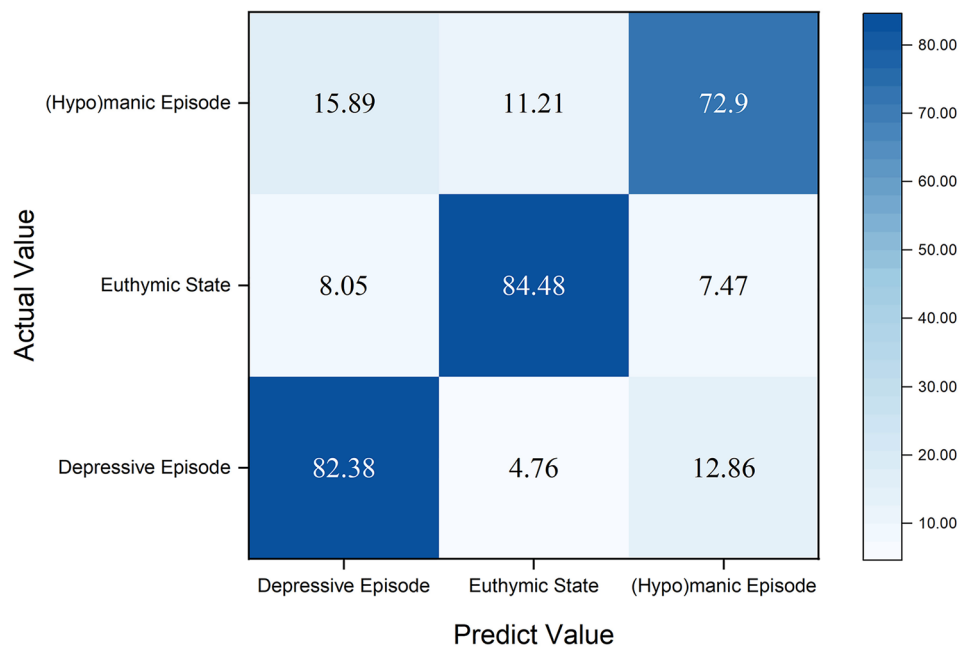


Fig. 4 LSTM model confusion matrix

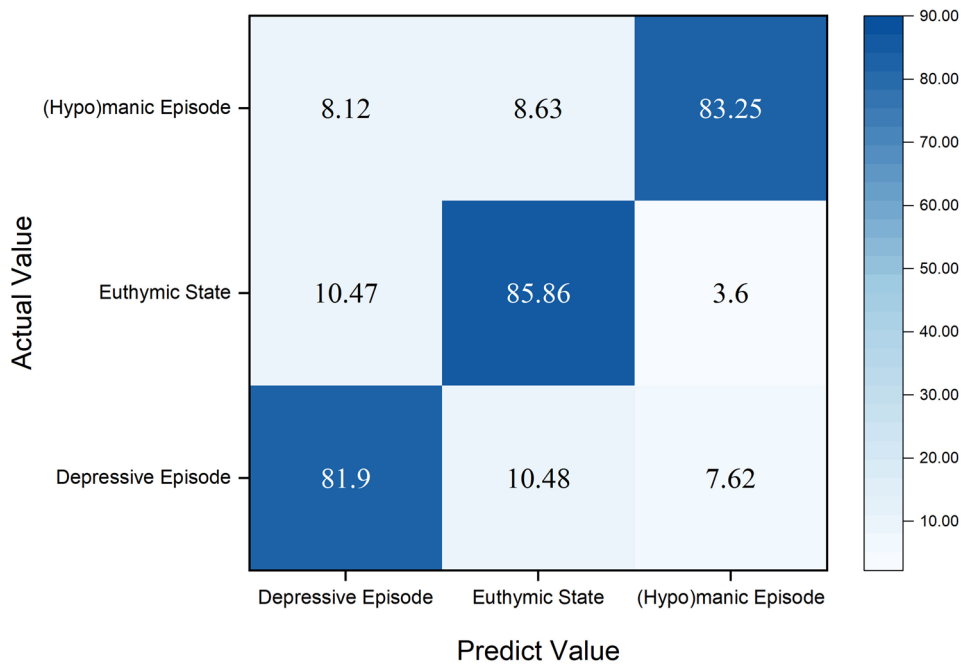


Fig. 5 ECAPA-TDNN model confusion matrix

ECAPA-TDNN model demonstrates classification accuracies exceeding 80% across all states: (hypo)manic episode (83.25%), depressive episode (81.90%), and euthymic state (85.86%).

Figure 6 demonstrates the confusion matrix for the (-) TFA+Ours model. The model achieved high recognition accuracy for both the depressive episode (90%) and the euthymic state (89.53%). However, there was a high

level of confusion between the (hypo) manic episode and the other two mood states, with 11.68% misclassified as depressive episode and 12.18% misclassified as euthymic state.

As shown in Fig. 7, the Ours model achieved recognition accuracies of 95.81% for the euthymic state, 90% for the depressive episode, and 78.18% for the (hypo) manic episode.

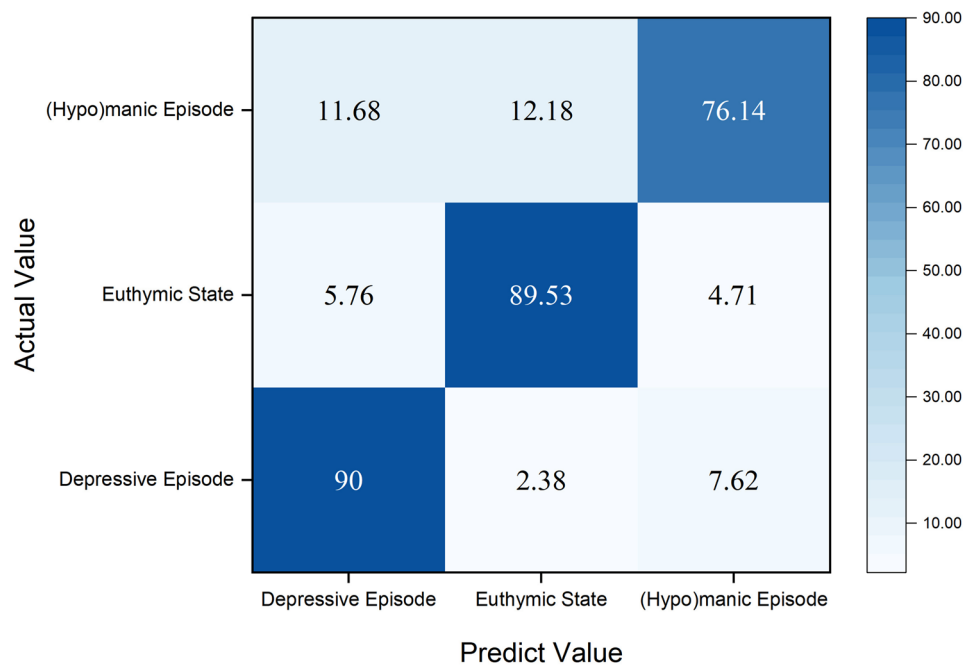


Fig. 6 (-)TFA+Ours model confusion matrix

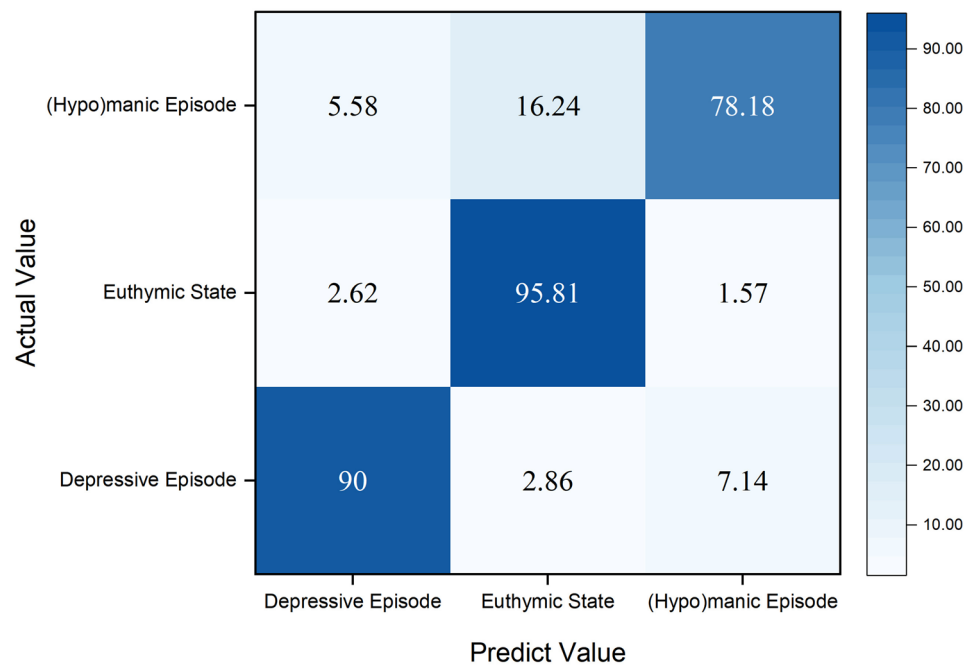


Fig. 7 Confusion matrix for the Ours model

Compared with LSTM model, the Ours model substantially increased recognition accuracy by 5.28% for the (hypo)manic episode, 7.62% for the depressive episode, and 11.33% for the euthymic state.

Compared with ECAPA-TDNN model, the identification accuracy improved by 8.10% for the (hypo)manic episode and 9.95% for the euthymic state, while

the improvement for the depressive episode was not significant.

Compared with (-)TFA+Ours model, the Ours model improved the recognition accuracy by 2.04% for the (hypo)manic episode and 6.28% for the euthymic state, while maintaining the same high accuracy (90%) for the depressive episode.

Discussion

This study developed a recognition model for the three mood states of bipolar disorder—(hypo)manic episode, depressive episode, and euthymic state—using Mandarin speech data. By integrating self-supervised feature extraction with deep learning algorithms, the Ours model outperformed baseline methods, achieving WA of 85.79%, UA of 85.94%, and Macro_F1 of 85.78%.

The study found that WavLM and HuBERT self-supervised features yielded higher speech recognition rates compared to traditional Mel-spectrogram features. This finding is consistent with previous studies on speech analysis [31, 32], which show that self-supervised feature extraction techniques can effectively learn deep representations of speech signals from unlabeled data, thereby enhancing performance across various speech tasks. Previous studies on speech recognition in bipolar disorder mostly relied on traditional acoustic features, such as fundamental frequency and derived parameters, to analyse speech feature differences across mood states [13, 17, 47]. However, the application of traditional speech features in bipolar disorder speech recognition has faced limitations in accuracy. For example, Karam et al. constructed a support vector machine (SVM) model using 52 traditional speech features, achieving 61% and 59% accuracy for manic and depressive episode recognition, respectively [17]. The use of self-supervised features in this study provides empirical evidence for the potential of self-supervised methods in speech analysis for mental disorders research. Furthermore, the superior performance of WavLM self-supervised features relative to HuBERT self-supervised features in this study may be attributed to WavLM's enhanced denoising ability. Moreover, WavLM uses gated relative position encoding in the Transformer encoder, which incorporates relative position information into attention network calculations to enable better modeling of local speech information.

Previous studies have applied deep learning to speech recognition in bipolar disorder [48–50]. The classification accuracy in these studies ranged from 60% to 72%, with an unweighted average recall rate between 45 and 65%. The Ours model in this study achieved stronger performance than prior studies, which may be due to the integration of the temporal modeling capabilities of LSTM, the multiscale feature aggregation of ECAPA-TDNN, and the key frequency band selection mechanism of TFA. In speech signal analysis, mood states within the same class usually exhibit a similar pattern of change in the time dimension. However, the change of different mood states in the time dimension shows significant variability, which greatly affects the role of temporal context information in the results of speech mood state recognition. Although the ECAPA-TDNN model can capture temporal context information through dilated convolution, it

has limitations in fully capturing temporal information. Therefore, based on the ECAPA-TDNN model, this study adds an LSTM layer before the SE_Res2Block in the model to capture the contextual information, and then passes the SE_Res2Block to enrich the mood states features. Since different frequency bands of the speech spectrum contribute differently to mood state recognition, Niu et al. proposed a Time-Frequency Attention (TFA) mechanism to obtain the timestamps and frequency bands related to mood state detection [51]. In this study, the above model is through Time-Frequency Attention again to construct the Ours model. The TFA layer in the Ours model identifies key timestamps and hidden representations (which encode multiple acoustic features) associated with mood states. For example, it highlights low-frequency bands (0.5–1 kHz) and slower timestamps (1.5–3s) in depressive episodes—consistent with clinical observations of reduced speech rate and lower pitch in depressed patients. In (hypo)manic episodes, the TFA layer emphasizes high-frequency bands (2–4 kHz) and faster timestamps (0.5–1.5s), aligning with increased speech rate and higher pitch in manic states. The TFA layer not only captures the temporal context but also enhances the model's interpretability by identifying the specific time intervals and embedding dimensions that are most relevant to mood state detection. This could provide significant clinical insights by allowing real-time monitoring of speech patterns.

The Ours model achieved higher recognition accuracy for the euthymic state and the depressive episode, though, some misclassifications persisted. One contributing factor may be attributed to the study's categorization of bipolar disorder into three categories, whereas the mood states in bipolar disorder exhibit a continuous spectrum [52]. Consequently, it is challenging for the recognition model to distinguish patients in transitional phases (e.g., hypomania and euthymia). Additionally, clinical symptoms such as anxiety might be partially associated with vocal alterations. Weiner et al. [53] employed acoustic features to identify mixed episodes in bipolar disorder and found that accurately classified cases had higher anxiety scores compared to misclassified cases. Furthermore, recognition accuracy for (hypo) manic episode remained lower than that for the other two states. This discrepancy may be due to the limited amount of mood-specific information conveyed in the speech of manic patients in this study, which might have increased the difficulty of model recognition.

This study demonstrates the potential of recognizing different mood states through the analysis of natural conversation audio samples based on deep learning algorithm. With further development, this model could be embedded in smartphone applications for remote monitoring, enabling continuous, real-time tracking of speech

patterns in patients with bipolar disorder. Such systems could provide clinicians with ongoing assessments and early warnings of mood shifts or impending episodes, facilitating timely interventions.

Although this study has made progress in speech-based recognition of mood states in bipolar disorder, there are several limitations. First, the sample size is relatively small, which may constrain external validity. Second, the recognition model proposed in this study did not include demographic and clinical variables, such as sex, age, or the type of medication. Their potential correlation with acoustic features cannot be entirely ruled out. For instance, some speech features used to differentiate mood states may be influenced by medication-related physiological effects on speech. Third, the proposed recognition model was primarily trained on a standard Mandarin, which means it may not fully capture the subtle acoustic features introduced by dialects and accents. This could potentially affect its recognition performance with speakers who have a strong regional accent. Finally, the absence of a healthy control group prevents assessment of discriminability between bipolar and typical speech, limiting generalizability.

To address these limitations, future studies should be conducted in a larger, more demographically balanced cohort and should control for potential confounds. Additionally, future research should integrate a more diverse corpus to improve the model's robustness when faced with various Mandarin accents and variations. Furthermore, follow-up studies should measure the changes in speech over the onset, course, and remission of bipolar disorder to investigate the causality and direction between speech and different mood states.

Conclusion

This study analyses the speech features of the three mood states in bipolar disorder, depressive episode, (hypo) manic episode, and euthymic state within the context of Mandarin speech. By systematically integrating speech analysis techniques into the identification process of bipolar disorder mood states, the research involves several critical steps, including speech data preprocessing, feature extraction, mood states recognition model construction, and analysis of recognition results, thereby establishing a comprehensive speech signal processing workflow. The Ours model in this study demonstrated better recognition performance, providing evidence for the use of speech parameters as biomarkers to accurately distinguish mood states in bipolar disorder.

Abbreviations

BRMS	Bech-Rafaelsen Mania Rating Scale
DSM-5	Diagnostic and Statistical Manual of Mental Disorder, Fifth Edition

ECAPA-TNDD	Emphasized Channel Attention, Propagation and Aggregation in TDNN
F0	Fundamental Frequency
FAST	Functioning Assessment Short Test
HAMD-17	17-item Hamilton Depression Rating Scale
IQR	Inter Quartile Range
LSTM	Long Short-Term Memory
MEL	Mel-spectrogram
RNNs	Recurrent Neural Networks
SCID-5-RV	The Structured Clinical Interview for DSM-5, Research Version
SE_Res2Block	Squeeze-and-Excitation Res2 Block
TDNN	Time Delay Neural Network
TFA	Time-Frequency Attention
UA	Unweighted Accuracy
WA	Weighted Accuracy

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Author contributions

Jinhao Li and Qiuping Yan recruited the participants, conducted clinical assessments, and wrote the manuscript; Minghui Li designed the study, conducted the data analysis, and revised the manuscript; Xiaofei Hou, Chao Li, and Ziping Zhao conducted speech signal preprocessing and speech feature extraction; Huifang Yin and Guangming Xu took the responsibilities for the integrity of data and the accuracy of data analysis, reviewed and edited the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of Tianjin Anding Hospital (date: 2023 March, approval number: 2023–07) and was conducted in accordance with the ethical standards set out in the 1964 Declaration of Helsinki and its subsequent amendments. All patients gave informed consent and signed the informed consent form.

Consent for publication

No applicable.

Competing interests

The authors declare no competing interests.

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